

Lifecycle Policy for storage oobjects

Tasks-1.Create a new storage bucket in your preferred cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) and upload 3 sample files named song.mp3, food.jpg, and bill.pdf, simulating files from apps like Spotify, Zomato, and Paytm.

? Open the [Amazon S3 Console](#).

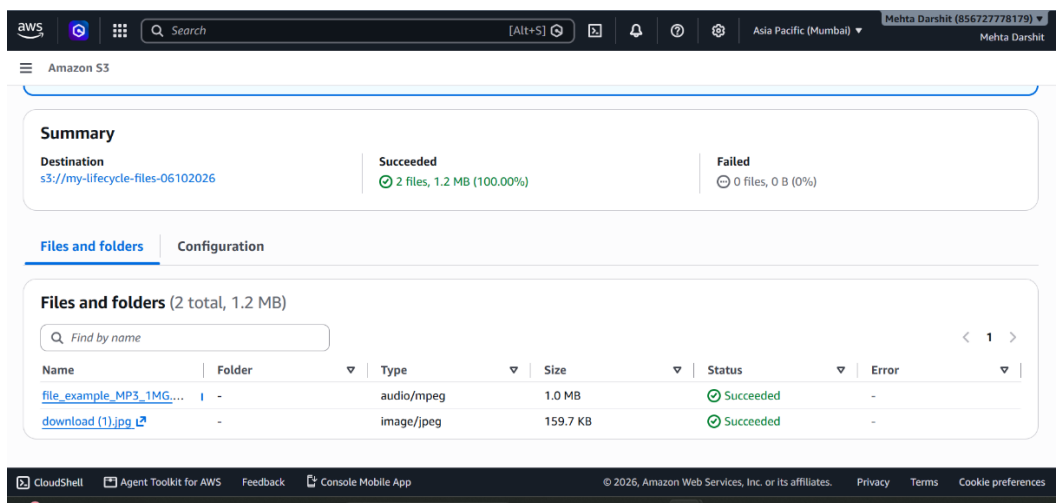
? Click Create bucket.

? Give it a unique name, for example:
my-lifecycle-files-20261005

? Select your required AWS region, such as Asia Pacific (Mumbai).

? Keep Block all public access enabled.

? Click Create bucket.



Tasks-2.Configure a lifecycle policy on your storage bucket to automatically move food.jpg to a cheaper storage class (like S3 Glacier, Google Coldline, or Azure Cool) after 1 day and delete bill.pdf after 3 days. Use your cloud provider's console or CLI to set up these rules, and double-check the transition and expiration actions.

Rule 1 — Move food.jpg to Glacier after 1 day

Go to:

S3 → your bucket → Management → Lifecycle rules → Create lifecycle rule

Set:

Rule name:

food-to-glacier

For the scope, choose **Limit the scope to a single prefix or tag**.

Because you want only food.jpg, the simplest option is to use the **object tag** approach or a prefix if you organize the object accordingly. If the object is literally at the bucket root as food.jpg, a prefix filter of food would also match objects beginning with food, so a tag is more precise.

For example, add the object tag:

Key: lifecycle

Value: food

to food.jpg, and configure the lifecycle rule to filter on that tag.

Under **Lifecycle rule actions**, select:

Transition current versions of objects between storage classes

Set:

After 1 day → Glacier Flexible Retrieval

Save the rule.

Rule 2 — Delete bill.pdf after 3 days

Create another lifecycle rule:

Rule name:

delete-bill

Use a filter that identifies bill.pdf, preferably a unique object tag such as:

Key: lifecycle

Value: bill

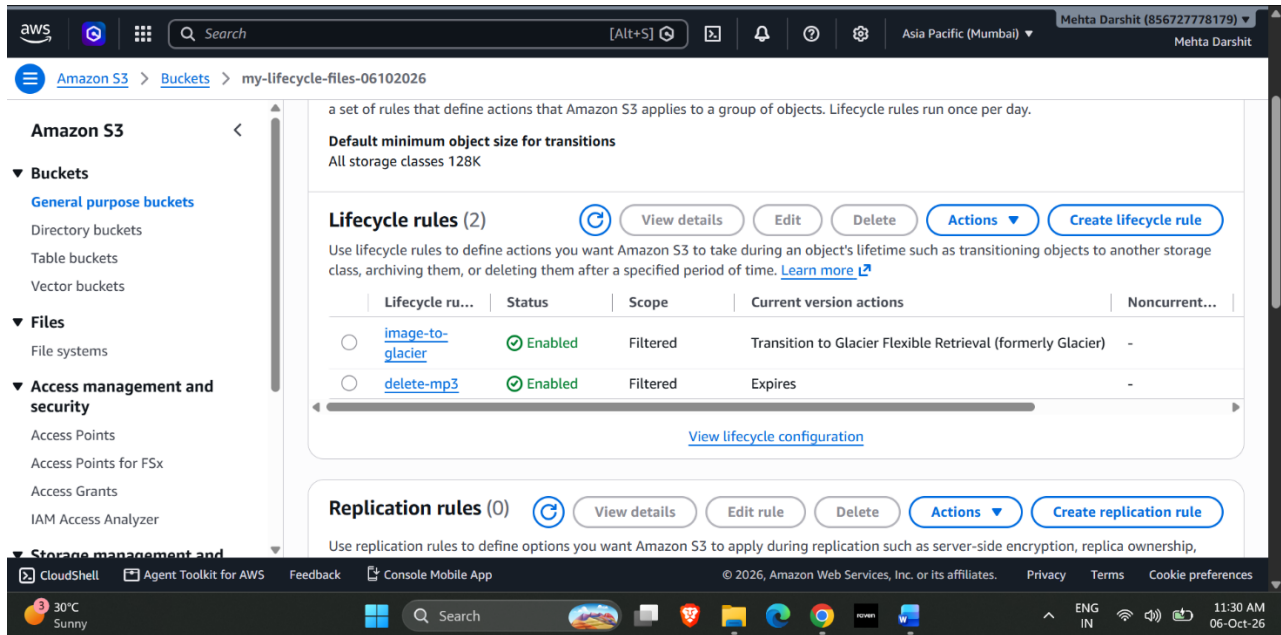
Then select:

Expire current versions of objects

Set:

3 days after object creation

Save the rule.



Tasks-3. Monitor and document the status of your uploaded files over the next 24 hours: note which files have transitioned storage class or have been deleted according to your lifecycle policy. Take a screenshot of the object details or use the CLI to confirm the changes.

File_example_mp3.mp3 → S3 Standard

downloads.jpg → S3 Standard

Tasks-4. Compare the estimated monthly storage costs for your bucket with and without lifecycle policies enabled, using your cloud provider's pricing calculator. Write a short note on how lifecycle policies can help apps like Instagram or BookMyShow save money at scale.

Scenario	Estimated monthly storage cost
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Without lifecycle policy	0.14USD
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With lifecycle policy	0.08 USD
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Estimated saving	0.06 USD
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Tasks-5. Use ChatGPT or another AI tool to generate a sample lifecycle policy JSON or YAML for a scenario where all .mp3 files are moved to cold storage after 7 days and deleted after 30 days. Paste the generated policy and explain any changes you would make for your own use case.

```
{
  "Rules": [
    {
      "ID": "MP3ColdStorageAndDeletion",
      "Status": "Enabled",
      "Filter": {
        "Suffix": ".mp3"
      },
      "Transitions": [
        {
          "Days": 7,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 30
      }
    }
  ]
}
```

Explanation

Rules

Contains the lifecycle rules that S3 should apply.

ID

MP3ColdStorageAndDeletion

Provides a descriptive name for the rule.

Status

Enabled

Activates the lifecycle rule.

Filter

```
"Filter": {  
  "Suffix": ".mp3"  
}
```

Limits the rule to objects whose names end in .mp3. Therefore, files such as:

song.mp3

music.mp3

podcast.mp3

are affected, while food.jpg and bill.pdf are not.

Transitions

```
"Transitions": [  
  {  
    "Days": 7,  
    "StorageClass": "GLACIER"  
  }  
]
```

Moves matching MP3 objects to the S3 Glacier storage class after 7 days.

Expiration

```
"Expiration": {  
  "Days": 30  
}
```

